

Heran J. Patel

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Education

The University of Texas at Austin - *Aerospace Engineering BS with Computational Engineering Certificate*, August 2019 - May 2023

- GPA: 3.41/4.00

The University of Texas at Austin - *Computer and Data Science - Artificial Intelligence MS*, January 2024 - May 2026

- GPA: 3.67/4.00

NVIDIA - *DLI Certifications*, June 2024 - September 2024

- Applications for AI for Predictive Maintenance, Building Transformer-Based Natural Language Processing Applications, Efficient Large, Language Model (LLM) Customization, Fundamentals of Accelerated Data Science, Fundamentals of Deep Learning.

Experience

Lockheed Martin, Huntsville, AL - *Software Engineer II*, June 2025 - Present

- Plan, conduct, and coordinate software development activities.
- Design, develop, document, test, and debug software that contains logical and mathematical solutions to mission problems or questions in computer language.
- Correct program errors, prepare operating instructions, compile documentations of program development, and analyze system capabilities to resolve questions of program intent, output requirements, input data acquisition, programming techniques, and controls.

Lockheed Martin, Fort Worth, TX - *Systems Engineer II*, June 2023 - June 2025

- Team Co-Lead in Model Based Systems Engineering (MBSE)
- Responsible for mission planning, ground security, and system/software requirements development, AGILE Software Development as well as performing MBSE tasks.
- Work in both a software and hardware environment.
- Experience creating IDD and ICDs and presenting at SRR, PDR, CDR

Quaternion Studios, College Station, TX - *Machine Learning Research Internship*, January 2023 - June 2023

- Alongside 2 other ML engineers, assisted them in researching and developing ML models from inception to deployment.
- Collaborated to transform business needs into technical solutions.
- Created and fine-tuned algorithms for processing vast datasets
- Enhanced model performance with fine-tuning

SpaceX, McGregor, TX - *Mentee for Independent Study and Mentorship Program*, January 2019 - April 2020

- Conducted research with SpaceX Aerospace Engineer, Jaymik Patel. Together we created a CAD Blended-Wing Body design for an airplane. Our goal was to reduce the carbon emissions of commercial aircraft by 8%-15%. Here is a [digital portfolio](#) documenting our journey and our [paper](#).

Projects

NASA MITTIC — *NASA, Houston, TX* - August 2022 - May 2023

Selected along with 4 others to represent UT Austin, we created a spinoff concept of a NASA IP for commercialization. We reinvented NASA's ZONE (Zeroing Out Negative Effects). Placed 7th nationally.

Predicting Bike-Sharing Demand Using Machine Learning and Weather Data — November 2025

Developed a production-ready machine learning pipeline that forecasts station-level bike share demand using engineered temporal, weather, and lag features, optimizing performance with ensemble methods and neural networks. I have also written a paper on this project that is currently being pushed for publication by The University of Texas at Austin.

Skills

Coding Languages: C++, Matlab, Java, Python.

Other: Atlassian Tools, AWS, JIRA, Confluence, Docker, Git, Github, Gitlab, GitKraken, Simulink, Linux, Web Designing, Microsoft Office/Suite, Capstone Diploma, Research License, NASA L'SPACE Academy Certificate, Machine Shop License, Jupyter Notebook, BitBucket, Agile/Scrum, Project Management, Cameo, Rhapsody, DOORS, SolidWorks, Visio, Autodesk Inventor, Autodesk Fusion 360.